

CS 774

July 29, 2025

Last / Pass. Logistics

Objective: What is Machine Learning?

- ① Learning Fundamentals of ML
- ② Specific ML models
- ③ Gets hands on experience on real data

What are you not?

- ↳ Research capability
- ↳ Learning Models / Bounds
- ↳ Strong foundation of mathematics of behind ML algorithms

① Research capability

↳ ML problems for real-world data

What is ML?

Same / Similar $\rightarrow$
- dist

1. Past Data

Predict (Future Data)

Findings Neural Patterns

① To find neural patterns in training data & make future prediction } Supervised Learning

② Φ $X \rightarrow Y \rightarrow$ output
 $\hookrightarrow$ All possible (instances)
 $\hookrightarrow h_0$ (hypothesis) $\in \mathcal{H}$

③ Learning to learn a program

$\{x^u, y^u\} \xrightarrow{\text{Data}} \boxed{\text{Apo}} \rightarrow (\underline{M} / \underline{Go})$

ho: $x \rightarrow y$